portable, cryptographically verifiable attestations issued by entities that actually know the facts: employers, certification bodies, customers, clinics, manufacturers, and others. Rather than ask whether ACME has a record for a specific person, the authorization system asks whether trusted issuers attest that the person has the attributes needed for the action they are attempting.

With this shift, ACME can grant access to contractors without creating new accounts, trust suppliers without setting up IdP integrations, recognize professional authority without maintaining directories of external certifications, and handle emergency access in a way that remains governed and auditable. The rest of this chapter explores how verifiable credentials work in practice and how they integrate with ACME's policy-based authorization architecture.

16.1 Why authorization needs verifiable credentials

As we discussed in chapter 12, reliable authorization depends on knowing who is taking an action, what they are permitted to do, and under what circumstances the action should be allowed. For internal users, ACME can draw these facts from authoritative internal systems such as HR records and the corporate directory. However, as ACME's collaboration platform has expanded, more and more authorization decisions involve people who have no accounts in these systems. Contractors, auditors, supplier representatives, clinicians, and field technicians all need controlled access to ACME's services. ACME does not employ these people, cannot manage their lifecycle events, and lacks authoritative knowledge of their roles and qualifications.

This reality challenges traditional approaches. Account-based models require every external person to be added to ACME's directory, even if they work for a different organization. This leads to identity records that ACME cannot verify and lifecycle responsibilities it cannot realistically manage. Federation solves part of the problem when both parties have the necessary infrastructure, but many of ACME's partners do not. Even when federation is feasible, it creates a fixed, bilateral relationship that is hard to scale across